

Autoencoder-Based Anomaly Detection for Breast Cancer Diagnosis

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Section 1: Problem Description and Dataset

Problem Description

Breast cancer diagnosis prior to the 1980s relied primarily on open surgical biopsies, a highly invasive procedure requiring general anesthesia and significant recovery time (Kim et al., 2025). The current standard of care has shifted to needle biopsies, which are minimally invasive and performed under local anesthesia (Dahabreh et al., 2014; *Why Would I Need a Breast Biopsy?*, n.d.). Fine-needle aspiration (FNA) was widely used for breast cancer diagnosis before being largely replaced by core needle biopsy (CNB), which retrieves a larger tissue sample and is considered more diagnostically reliable today (Kim et al., 2025). Both methods allow a pathologist to examine tumor tissue and determine whether a mass is benign or malignant. However, FNA is still used in low to middle income developing countries as it is cost effective and largely accurate (Kim et al., 2025; Rubin et al., 1997), with studies finding 100% positive predictive value, 100% specificity and 87% sensitivity (Rubin et al., 1997). In addition, pathological diagnosis remains subject to variability among different pathologists examining the same sample which can affect diagnostic consistency and patient outcomes.

To reduce subjectivity, increase diagnostic speed, and improve accuracy, this analysis applies an autoencoder, to image processing features derived from FNA biopsies. The goal is to apply this unsupervised machine learning technique to standardize areas of diagnostic ambiguity, flag samples that need closer review, and ultimately reduce turnaround time so that treatment can begin sooner.

Dataset

The dataset used in this analysis is the Wisconsin Breast Cancer Diagnostic dataset, comprised of 569 patients (William Wolberg, 1993). For each patient, 30 features were computed from a digitized image of their fine-needle aspirate biopsy of a breast mass. Ten characteristics were measured per cell nucleus, including radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension. Each characteristic is summarized three ways: by mean value, standard error, and worst (largest) value across the cells present in the image. This yields $10 \times 3 = 30$ features per patient. All 30 features are continuous. Each patient is also associated with a diagnosis label: benign or malignant. This label is not used during autoencoder training, as the method is unsupervised; However, the model is only presented benign sample data during training. Of the 569 patients, 357 are diagnosed as benign and 212 as malignant.

Section 2: ML Algorithm: Autoencoder

Overview

An autoencoder is a type of neural network that uses unsupervised machine learning to compress and reconstruct input data to learn important underlying patterns. Rather than predicting a label, autoencoders are trained to discover latent variables, the underlying structure of the data that is not directly observable. Latent variables represent a transformation of high-dimensional data into a lower-dimensional space, describing the data in a simpler more compact form.

Applications of autoencoders include noise removal, data quality improvement, anomaly detection, and feature extraction. In this analysis, the autoencoder is used for anomaly detection. Samples with unusually high reconstruction error are flagged as potentially malignant because the malignant tissue deviates from the benign patterns learned during training.

Why an autoencoder is appropriate

The autoencoder is suited for this problem for several reasons. First, all 30 features in this dataset are continuous numerical measurements, which is well suited for an autoencoder trained with MSE loss, as MSE requires data that exists on a continuous scale to meaningfully quantify reconstruction error. Second, autoencoders assume that the high-dimensional data contains redundancy, this assumption is reasonable here, since many of the 30 features are mathematically related. Third, autoencoders perform well when the goal is to learn the structure of a dominant class, benign tumors in this case, and flag deviations from that structure. By training on the full dataset in an unsupervised fashion, the autoencoder learns to reconstruct typical cell nuclei features well, and samples that reconstruct poorly may warrant closer clinical attention.

How the algorithm works

There are three main components to an autoencoder: the encoder, the bottleneck, and the decoder.

The encoder compresses the raw input data into a smaller, more manageable form by reducing its dimensionality. It consists of successive hidden layers that progressively condense the representation. The bottleneck is the smallest layer in the network, where the data is most compressed, filtering out noise and redundancy and leaving only the essential structure. The decoder then takes this compressed representation and attempts to reconstruct the original input, using hidden layers that progressively expand the data back to its original dimensions. The quality of the reconstruction depends on how well the encoder-decoder pair learns to minimize the difference between the input and the output during training.

Since all 30 features in this dataset are continuous, Mean Squared Error (MSE) is used as the loss function. If the data contained binary features, Binary Cross-Entropy would be used instead.

Mathematical formulation

In mathematical terms, the encoder is denoted by G_ϕ , the bottleneck by z , and the decoder by F_θ . The encoder maps the high-dimensional input $x \in \mathbb{R}$ to a lower-dimensional latent vector z :

$$z = G_\phi(x)$$

The encoder function $z = G_\phi(x)$ is computed by passing the input through successive hidden layers, where each layer applies an affine transformation followed by a non-linear activation function:

$$h^{(l)} = \sigma(W^{(l)}h^{(l-1)} + b^{(l)})$$

An affine transformation scales and shifts the data; it multiplies the previous layer's output $h^{(l-1)}$ by a learned weight matrix $W^{(l)}$ and adds a bias vector $b^{(l)}$. The non-linear activation function σ , typically ReLu, is applied on top, allowing the network to capture complex non-linear patterns. The parameters ϕ refer collectively to all these learned weight matrices $W^{(l)}$ and bias vectors $b^{(l)}$ across every layer of the encoder.

The bottleneck z is the output of the encoder's final layer. This is the most compressed representation of the input, retaining only what the network has determined to be essential.

The decoder then reconstructs the input from z :

$$\hat{x} = F_\theta(z)$$

where $\hat{x}$ is the reconstructed output and θ refers collectively to all weights and biases in the decoder. The decoder mirrors the encoder in structure, expanding the compressed representation back to its original dimensions.

The network is trained by minimizing the MSE reconstruction loss across all n training samples:

$$\mathcal{L}(\theta, \phi) = \frac{1}{n} \sum_{i=1}^n \left| x^{(i)} - F_\theta \left(G_\phi(x^{(i)}) \right) \right|^2$$

This loss measures the average squared difference between each original input and its reconstruction. It is minimized through backpropagation, where the error is passed backward through the network and used to update all weights via gradient descent:

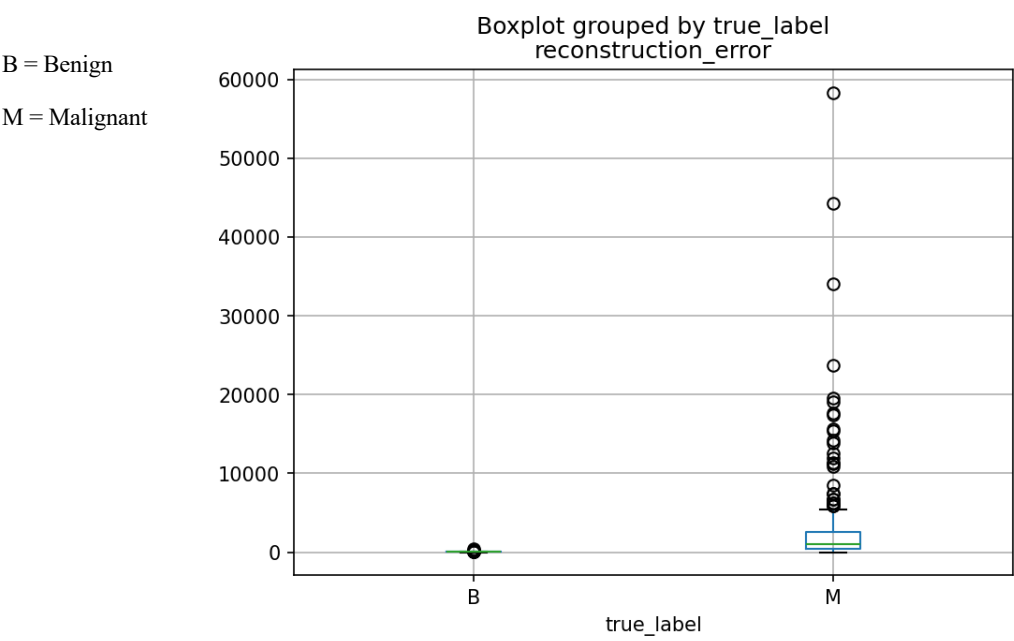
$$\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}, \quad \phi \leftarrow \phi - \eta \nabla_\phi \mathcal{L}$$

where η is the learning rate, controlling the size of each update step. Because the encoder and decoder are trained jointly, the network simultaneously learns how to compress and how to reconstruct, improving both until the reconstruction loss is minimized.

Section 3: Results

An autoencoder with three layers was constructed: an input layer accepting the 30 continuous features, a single dense encoding layer with 16 neurons and ReLU activation, and a dense decoding layer reconstructing the original 30 features using linear activation. The model was compiled with an Adam optimizer and Mean Squared Error (MSE) loss, with MSE chosen because all input features are continuous. The architecture was intentionally kept simple given the small training set of 285 benign samples, as a deeper network would risk overfitting.

Figure 1. Reconstruction Error Across Benign and Malignant Samples



The autoencoder was trained exclusively on benign tumor samples and tested on both benign and malignant tumors. The rationale for this approach is that a model trained only on benign phenotypes will reconstruct benign samples well but struggle to reconstruct malignant samples, due to their differential cellular characteristics, resulting in a higher reconstruction error for malignant cases. This behavior is confirmed in Figure 1, which shows the reconstruction error grouped by true diagnosis. Benign tumors cluster tightly near zero reconstruction error, while malignant tumors exhibit higher and more variable reconstruction error, with many outliers extending well above 10,000.

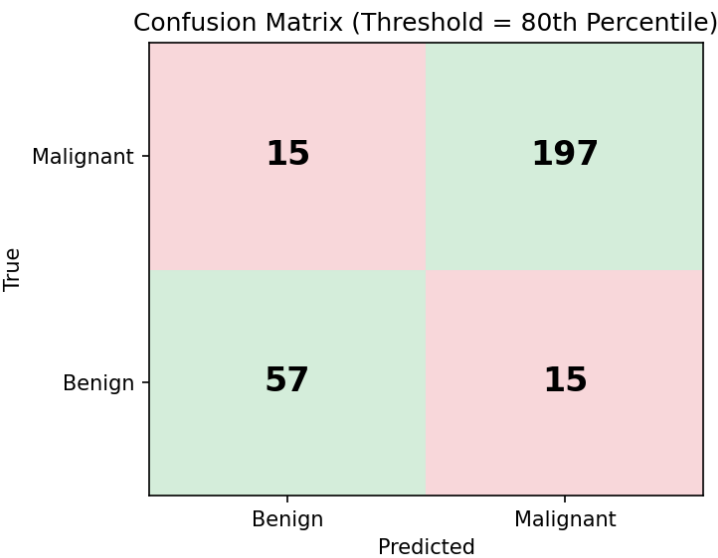
When evaluated on the full test set, the model achieved an Area Under the Curve (AUC) of 0.951, indicating strong discriminative ability between benign and malignant tumors. To generate classification predictions, a reconstruction error threshold was applied. Samples exceeding the threshold were flagged as malignant. Given the clinical context, minimizing false negatives

(missed malignancies) was prioritized over minimizing false positives, as failing to detect a malignant tumor carries far greater risk.

Table 1. Threshold Percentiles Comparisons

Percentile	Threshold	M-Recall	M-Precision	B-Recall	B-Precision	Accuracy	Missed Malignant
95th	123.34	0.877	0.979	0.944	0.723	0.894	26
90th	100.36	0.882	0.959	0.889	0.719	0.884	25
80th	60.4	0.929	0.929	0.792	0.792	0.894	15
70th	44.99	0.948	0.901	0.694	0.82	0.884	11

Figure 2. Confusion Matrix at 80th Percentile Threshold



At the 95th percentile threshold, the model achieved a precision of 0.72 and a recall of 0.94 but produced 26 missed malignant cases (Table 1). Table 1 illustrates the tradeoff between lowering the threshold to increase sensitivity at the cost of decreasing specificity. The 80th percentile threshold was selected as the optimal balance, reducing missed malignancies to 15 while maintaining an overall accuracy of 0.894. As shown in Figure 2, at this threshold the model correctly identified 197 of 212 malignant tumors and 57 of 72 benign tumors in the test set, where the 72 benign test samples reflect an 80/20 split of the 357 benign cases. The model achieved a precision and recall of 0.79 for benign tumors and 0.93 for malignant tumors, reflecting the intentional compromise made given the clinical screening context. It should be noted that this analysis was conducted on a relatively small cohort of 569 patients, and the model's performance should be validated on a larger, independent dataset before drawing broader clinical conclusions.

Works Cited

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